

Activity 14

The poor cartographer—Graph coloring

Summary

Many optimization problems involve situations where certain events cannot occur at the same time, or where certain members of a set of objects cannot be adjacent. For example, anyone who has tried to timetable classes or meetings will have encountered the problem of satisfying the constraints on all the people involved. Many of these difficulties are crystallized in the map coloring problem, in which colors must be chosen for countries on a map in a way that makes bordering countries different colors. This activity is about that problem.

Curriculum Links

- ✓ Mathematics: Number – Exploring numbers in other bases. Representing numbers in base two.
- ✓ Mathematics: Algebra – Continue a sequential pattern, and describe a rule for this pattern. Patterns and relationships in powers of two.

Skills

- ✓ Problem solving.
- ✓ Logical reasoning.
- ✓ Algorithmic procedures and complexity.
- ✓ Communication of insights.

Ages

- ✓ 7 and up

Materials

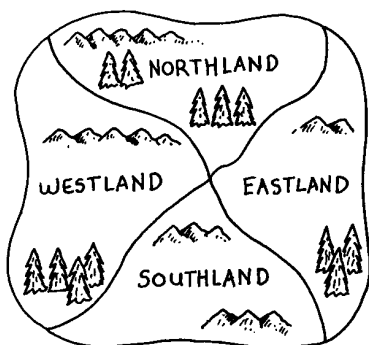
- ✓ a whiteboard or similar writing surface.
Each student will need:
- ✓ a copy of one or more of the worksheets,
- ✓ movable small colored markers (e.g. counters or poker chips), and
- ✓ four crayons of different colors (or colored pencils, felt tips etc.)

Graph Coloring



Introduction

This activity revolves around a story in which the students have been asked to help out a cartographer, or map-maker, who is coloring in the countries on a map. It doesn't matter which color a country is, so long as it's different to all bordering countries.



For example, this map shows four countries. If we color Northland red, then Westland and Eastland cannot be red, since their border with Northland would be hard to see. We could color Westland green, and it is also acceptable to color Eastland green because it does not share a border with Westland. (If two countries meet only at a single point, they do not count as sharing a border and hence can be made the same color.) Southland can be colored red, and we end up needing only two colors for the map.

In our story, the cartographer is poor and can't afford many crayons, so the idea is to use as few colors as possible.

Discussion

Describe the problem that the students will be working on, demonstrating the coloring process on a blackboard.

Give out a copy of the first worksheet. This map can be colored correctly using only two colors. Although restricting the number of colors to just two might sound particularly challenging, the task is quite simple compared with maps that require more colors because there is very little choice about what color each country can be.

Have the students try to color the map in with only two colors. In the process they may discover the “has-to-be” rule: once one country is colored in, any bordering country has to be the opposite color. This rule is applied repeatedly until all countries are colored in. It is best if the students can discover this rule for themselves, rather than being told it, as it will give them a better insight into the process.

As students complete each exercise they can be given the next sheet to try.

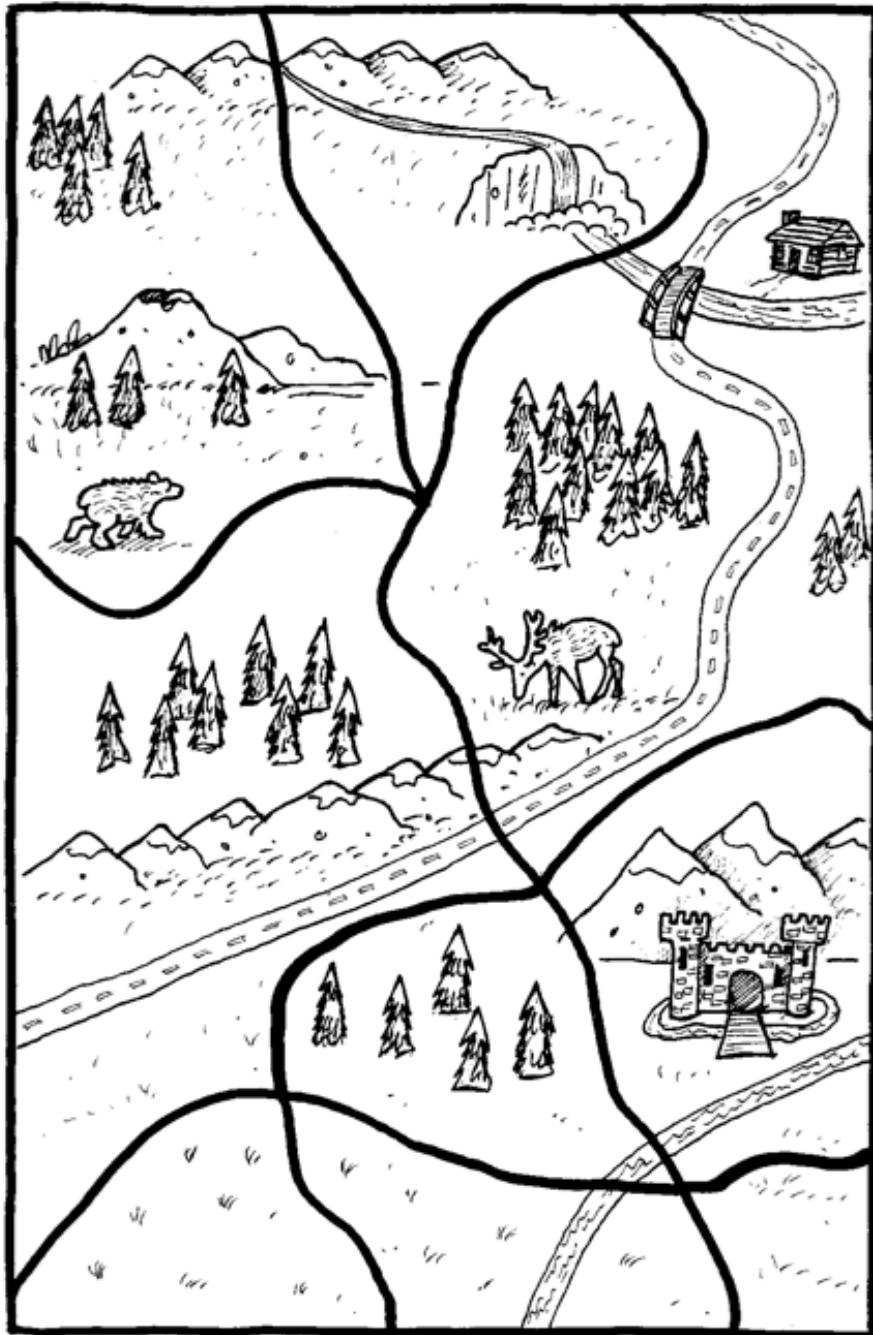
The students may also discover that it is better to use place-holders, such as colored counters, instead of coloring the countries straight away, since this makes it easier for them to change their mind.

For older students, ask them to explain how they know that they have found the minimum number of colors. For example, at least three colors are required for this map because it includes a group of three countries (the largest three), each of which has borders with the other two.

If a student finishes all the sheets early, ask them to try to devise a map that requires five different colors. It has been proved that any map can be colored with only four colors, so this task will keep them occupied for some time! In our experience students will quickly find maps that they believe require five colors, but of course it is always possible to find a four-color solution to their maps.

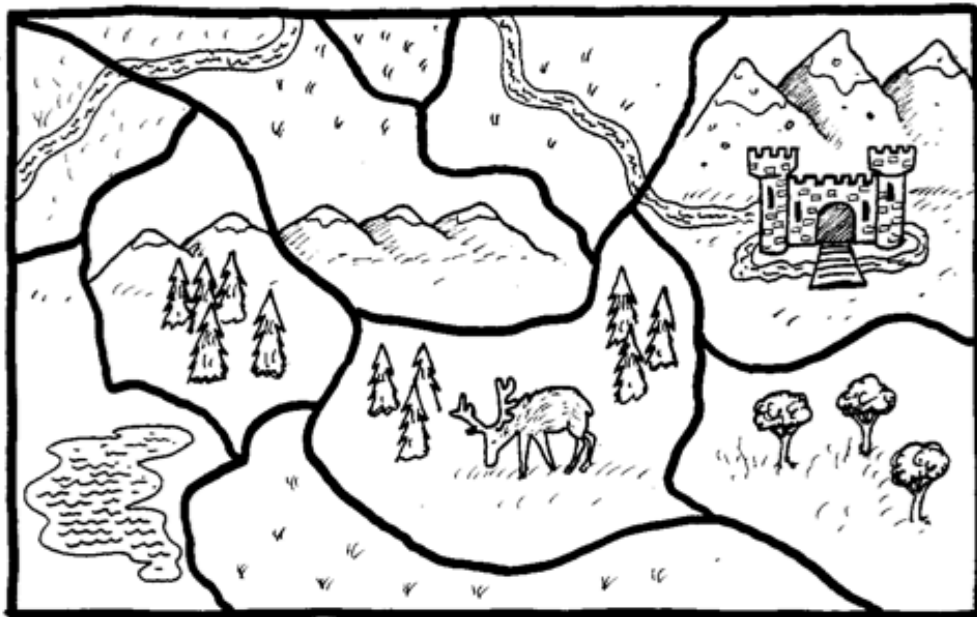
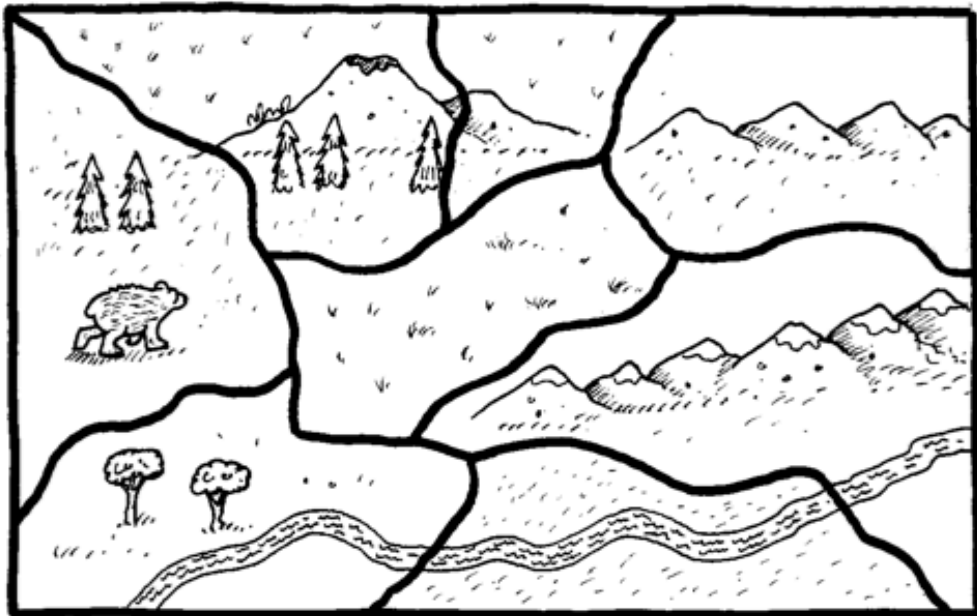
Worksheet Activity: Graph Coloring 1

Color in the countries on this map with as few colors as possible, but make sure that no two bordering countries are the same color.



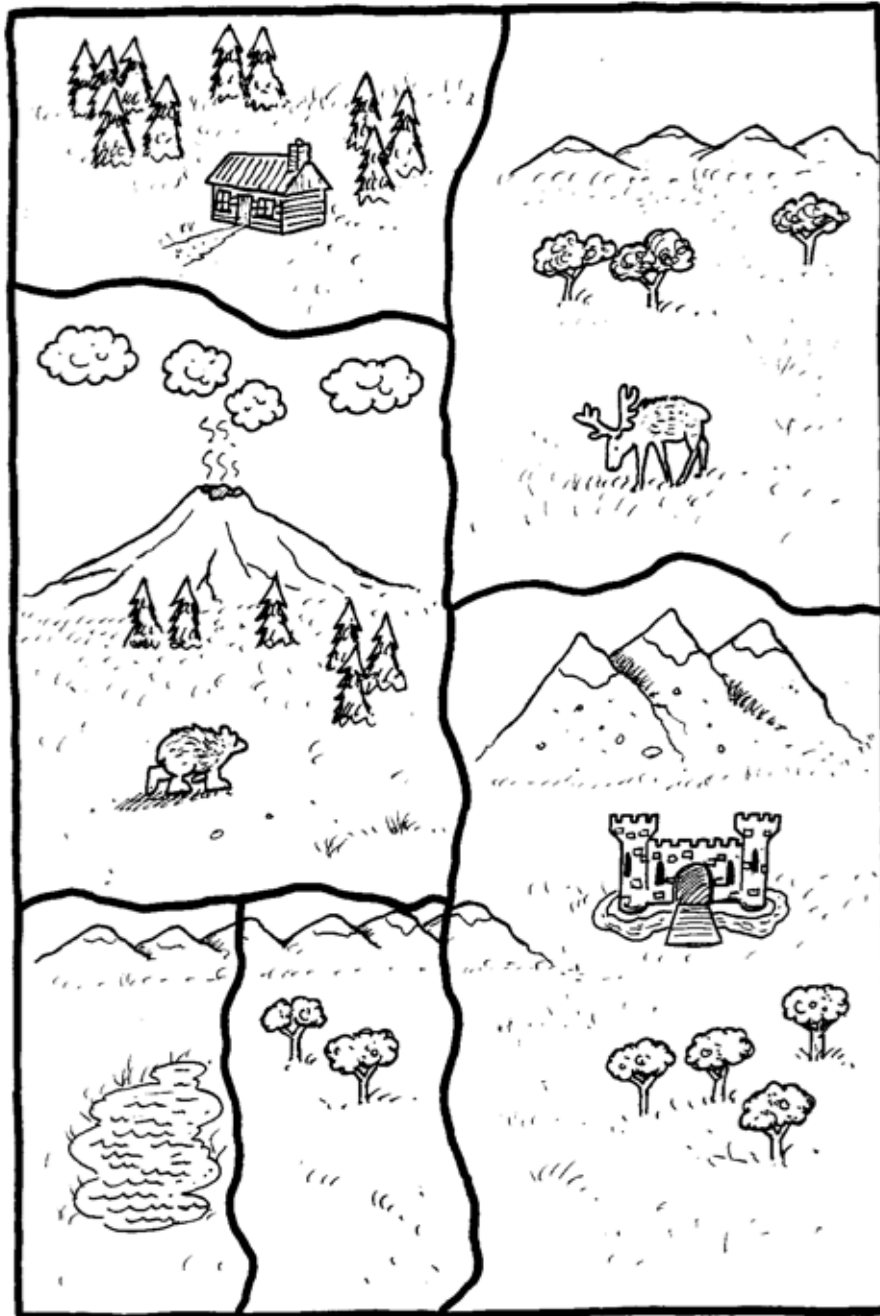
Worksheet Activity: Graph Coloring 2

Color in the countries on this map with as few colors as possible, but make sure that no two bordering countries are the same color.



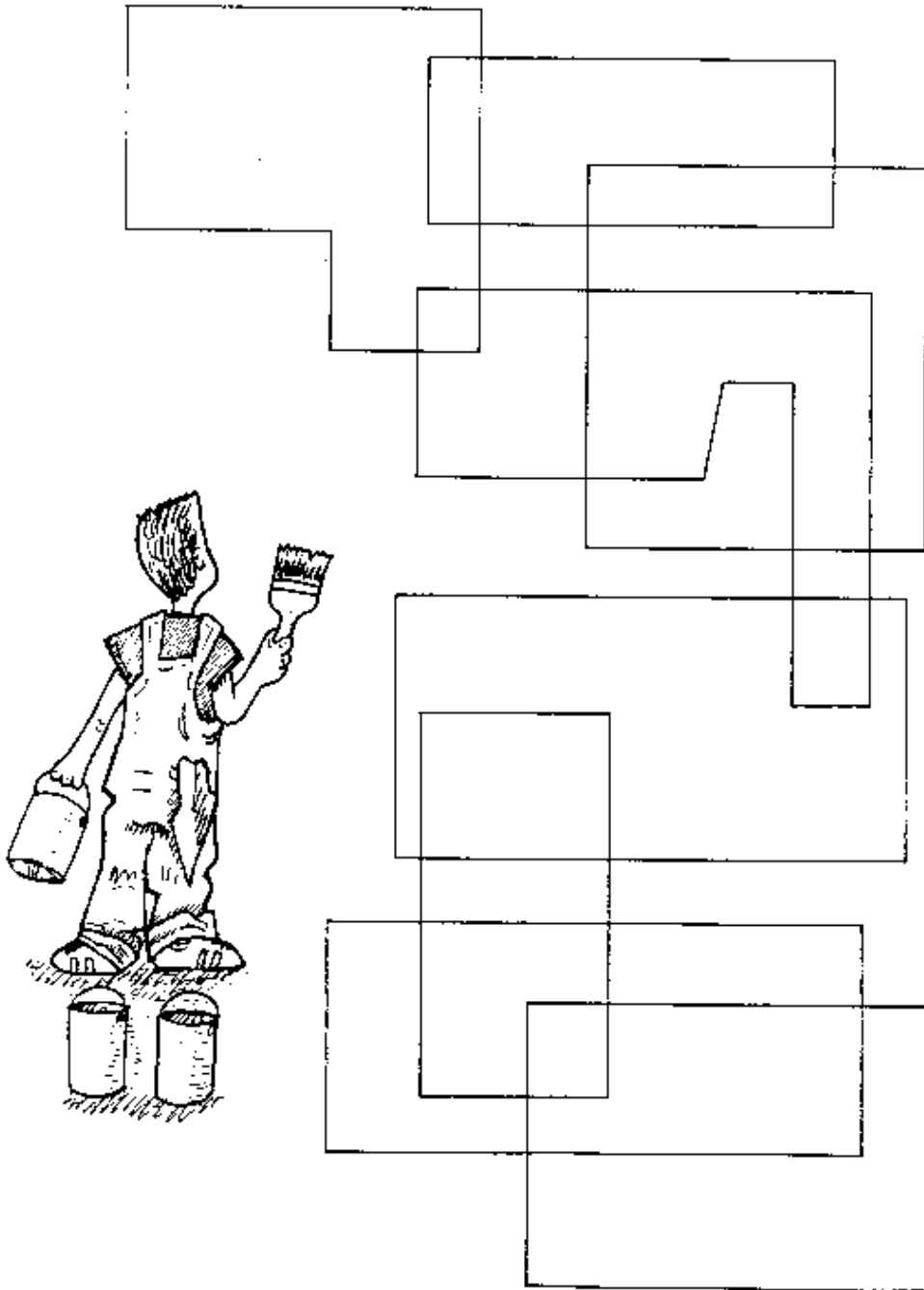
Worksheet Activity: Graph Coloring 3

Color in the countries on this map with as few colors as possible, but make sure that no two bordering countries are the same color.



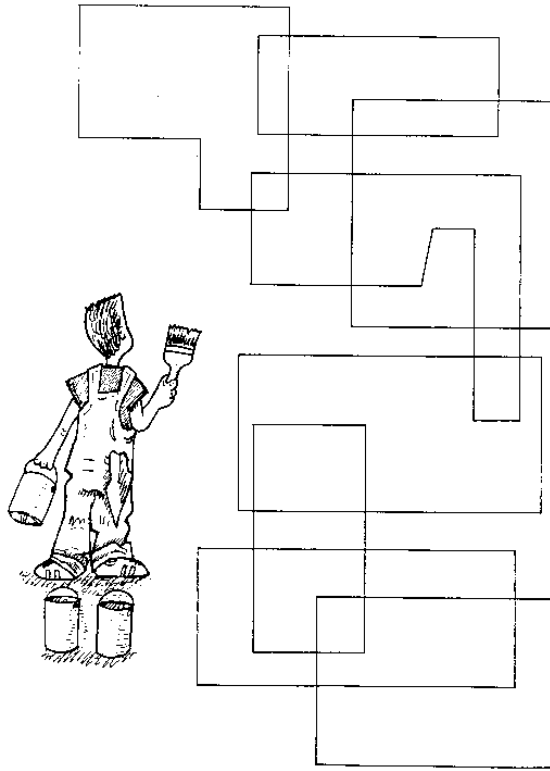
Worksheet Activity: Graph Coloring 4

Color in the countries on this map with as few colors as possible, but make sure that no two bordering countries are the same color.



Variations and Extensions

There is a simple way to construct maps that require only two colors, as shown here. This map was drawn by overlaying closed curves (lines whose beginning joins up with their end). You can draw any number of these curves, of any shape, on top of each other, and you will always end up with a map that can be colored with two colors. Students can experiment with creating this type of map.



Four colors are always enough to color a map drawn on a sheet of paper or on a sphere (that is, a globe). One might wonder (as scientists are paid to do) how many colors are needed for maps drawn on weirder surfaces, such as the torus (the shape of a donut). In this case, one might need five colors, and five is always enough. Students might like to experiment with this.

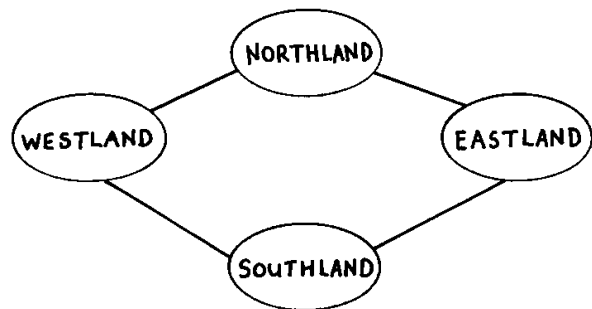
There are many other entertaining variations on the map-coloring problem that lead off into directions where much is currently unknown. For example, if I am coloring a map on a sheet of paper by myself, then I know that if I work cleverly, four colors will be enough. But suppose that instead of working alone I am working with an incompetent (or even adversarial) partner, and we take turns at choosing the color for countries. Assume that I work cleverly, while my partner only works “legally” as we take turns coloring countries on the map. How many crayons need to be on the table in order for me in my cleverness to be able to make up for my partner's legal but not very bright (or even subversive) moves? The maximum number isn't known! In 1992 it was proved that 33 crayons will always be enough, and in 2008 this was improved by a proof that 17 would be sufficient, but we still don't know that this many is ever actually required. (Experts conjecture that fewer than 10 colors are sufficient.) Students might enjoy acting out this situation, which can be played as a two-person game where you try to maximise the number of colours that your opponent needs.

In another variation of map coloring known as *empire coloring*, we start with two different maps on two sheets of paper having the same number of countries. Each country on one of the maps (say, the Earth) is paired with exactly one country on the other map (which might be colonies on the Moon). In addition to the usual coloring requirement of different colors for countries that share a border (for both maps) we add the requirement that each Earth country must be colored the same as its colony on the Moon. How many colors do we need for this problem? The answer is currently unknown.

What's it all about?

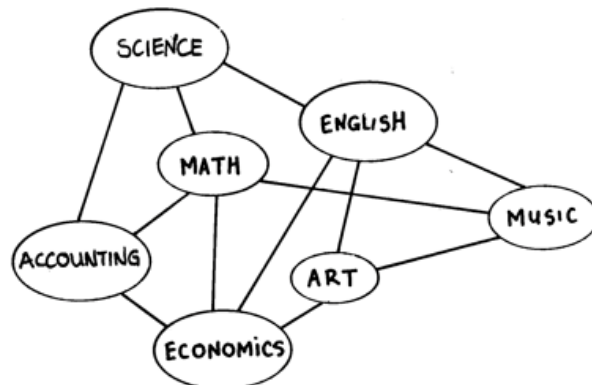
The map coloring problem that we have explored in this activity is essentially to find the minimum number of colors—two, three, or four—that are necessary to color a particular map. The conjecture that any map can be colored using only four colors was formulated in 1852, but it was not proved until 1976. Computer science is full of unsolved problems, and knowing that the four-color theorem was proved after more than 120 years of attention from researchers is an encouragement to those working on other problems whose solution has eluded them for decades.

Map coloring belongs to a general class of problems known as “graph coloring.” In computer science, a graph is an abstract representation of relationships, as shown here.



As mentioned in Activity 9 on the Muddy City, the term graph is used in a different sense in mathematics to mean a chart displaying numerical data, such as a bar graph, but the graphs that computer scientists use are not related to these. In computer science, graphs are drawn using circles or large dots, technically called “nodes,” to denote objects, with lines between them to indicate some sort of relationship between the objects. The above graph happens to represent the map at the beginning of this activity. The nodes represent the countries, and a line between two nodes indicates that those two countries share a common border. On the graph, the coloring rule is that no connected nodes should be allocated the same color. Unlike a map, there is no limit to the number of colors that a general graph may require, because many different constraints may be drawn in as connecting lines, whereas the two-dimensional nature of maps restricts the possible arrangements. The “graph coloring problem” is to find the minimum number of colors that are needed for a particular graph.

In the graph on the right the nodes correspond to subjects in a school. A line between two subjects indicates that at least one student is taking both subjects, and so they should not be timetabled for the same



period. Using this representation, the problem of finding a workable timetable using the minimum number of periods is equivalent to the coloring problem, where the different colors correspond to different periods. Graph coloring algorithms are of great interest in computer science, and are used for many real-world problems, although they are probably never used to color in maps!—our poor cartographer is just a fiction.

There are literally thousands of other problems based on graphs. Some are described elsewhere in this book, such as the minimal spanning tree of Activity 9 and the dominating sets of Activity 14. Graphs are a very general way of representing data and can be used to represent all sorts of situations, such as a map made up of roads and intersections, connections between atoms in a molecule, paths that messages can take through a computer network, connections between components on a printed circuit board, and relationships between the tasks required to carry out a large project. For this reason, problems involving graph representations have long fascinated computer scientists.

Many of these problems are very difficult—not difficult conceptually, but difficult because they take a long time to solve. For example, to determine the most efficient solution for a graph coloring problem of moderate size—such as finding the best way to timetable a school with thirty teachers and 800 students—could take years, even centuries, for a computer using the best known algorithm. The problem would be irrelevant by the time the solution was found—and that’s assuming the computer doesn’t break down or wear out before it finishes! Such problems are only solved in practice because we are content to work with sub-optimal, but still very good, solutions. If we were to insist on being able to guarantee that the solution found was the very best one, the problem would be completely intractable.

The amount of computer time needed to solve coloring problems increases exponentially with the size of the graph. Consider the map coloring problem. It can be solved by trying out all possible ways to color the map. We know that at most four colors are required, so we need to evaluate every combination of assigning the four colors to the countries. If there are n countries, there are 4^n combinations. This number grows very rapidly: every country that is added multiplies the number of combinations by four, and hence quadruples the solution time. Even if a computer were invented that could solve the problem for, say, fifty countries in just one hour, adding one more country would require four hours, and we would only need to add ten more countries to make the computer take over a year to find the solution. This kind of problem won’t go away just because we keep inventing faster and faster computers!

Graph coloring is a good example of a problem whose solution time grows exponentially. For very simple instances of the problem, such as the small maps used in this activity, it is quite easy to find the optimal solution, but as soon as the number of countries increases beyond about ten, the problem becomes very difficult to do by hand, and with a hundred or more countries, even a computer can take many years to try out all the possible ways of coloring the map in order to choose the optimal one.

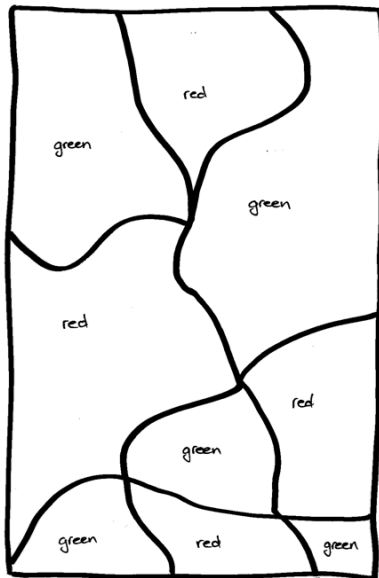
Many real-life problems are like this, but must be solved anyway. Computer scientists use methods that give good, but not perfect, answers. These *heuristic* techniques are often very close to optimal, very fast to compute, and give answers that are close enough for all practical purposes. Schools can tolerate using one more classroom than would be needed if the timetable were perfect, and perhaps the poor cartographer could afford an extra color even though it is not strictly necessary.

No-one has proved that there isn't an efficient way to solve this sort of problem on conventional computers, but neither has anyone proved that there is, and computer scientists are sceptical that an efficient method will ever be found. We will learn more about this kind of problem in the next two activities.

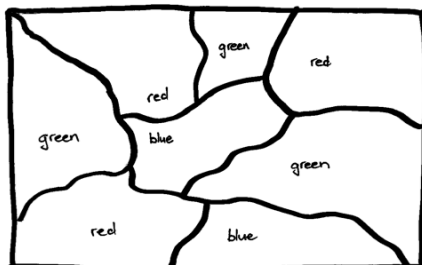
Further reading

Harel discusses the four-color theorem, including its history, in *Algorithmics*. More aspects of the map-coloring problem are discussed in *This is MEGA-Mathematics!* by Casey and Fellows. Kubale's 2004 book, *Graph Colorings*, includes a history of the problem. There are many websites that cover this topic.

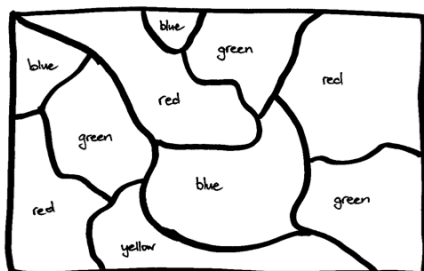
Solutions and Hints

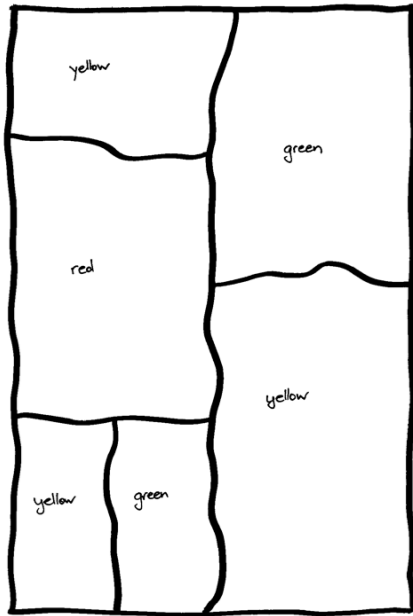


This is the only possible solution for the map on worksheet 1 (of course, the choice of colors is up to the student, but only two different colors are required).

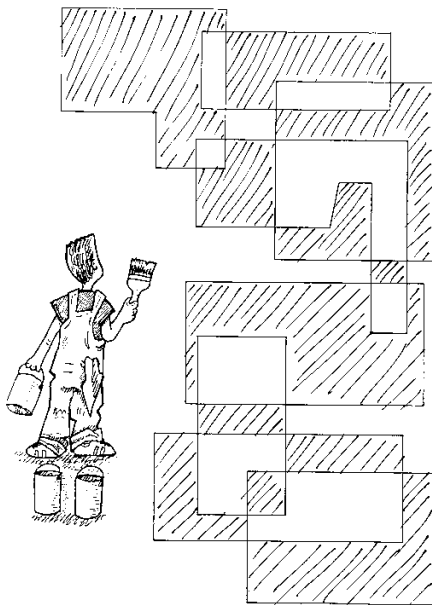


The map at the top of worksheet 2 can be colored correctly using three colors, while the one at the bottom requires four. Here are two possible solutions.





The map on worksheet 3 is a simpler three-color map, with a possible solution shown here.



Solution for worksheet 4 using just two colors (shaded and white).